



Boeing Bird of Prey

The **Boeing Bird of Prey** is an American black project aircraft, intended to demonstrate stealth technology. It was developed by McDonnell Douglas and Boeing in the 1990s.^[1] The company provided \$67 million of funding for the project;^[1] it was a low-cost program compared to many other programs of similar scale. It developed technology and materials which would later be used on Boeing's X-45 unmanned combat air vehicle. As an internal project, this aircraft was not given an X-plane designation. There are no public plans to make this a production aircraft. It is characterized as a technology demonstrator.

Design and development

Development of the Bird of Prey began in 1992 by McDonnell Douglas's Phantom Works division for special projects, at Area 51. The aircraft's name is a reference to the Klingon Bird-of-Prey, a fictional class of starships seen in the science fiction franchise Star Trek.^[2] Phantom Works later became part of Boeing Integrated Defense Systems after the Boeing–McDonnell Douglas merger in 1997.

The first flight was in 1996, and 39 more flights were performed through the program's conclusion in 1999.^[1] The Bird of Prey was designed to prevent shadows and is believed to have been used to test active camouflage, which would involve its surfaces changing color or luminosity to match the surroundings.^[3]

Because it was a demonstration aircraft, the Bird of Prey used a commercial off-the-shelf turbofan engine and manual hydraulic controls rather than fly-by-wire. This shortened the development time and greatly reduced its cost. (A production aircraft would have computerized controls.)

The shape is aerodynamically stable enough to be flown without computer correction. Its aerodynamic stability is in part due to lift provided by the chines, as used in other aircraft including the SR-71 Blackbird. This provided lift for the nose in flight. This configuration, which can be stable without a

Bird of Prey



General information

Type	Experimental stealth testbed
Manufacturer	<u>McDonnell Douglas</u> / <u>Boeing</u>
Status	On display
Number built	1

History

First flight	September 11, 1996
Retired	April 1999
Preserved at	<u>National Museum of the United States Air Force</u>



Boeing Bird of Prey concept art

horizontal tailplane and a conventional vertical rudder, is now a standard in later stealth unmanned aerial vehicles such as the X-45 and X-47, tailless aircraft which use drag rudders (asymmetrically used wingtip airbrakes) for yaw control.

The aircraft, which had given the designation "YF-118G" as a cover,^[4] was made public on October 18, 2002.^[1]

Aircraft on display

The Bird of Prey was put on display at the National Museum of the United States Air Force at Wright-Patterson Air Force Base near Dayton, Ohio on July 16, 2003. It is now on display at the museum's Modern Flight Gallery above their F-22 Raptor.



Boeing Bird of Prey at the National Museum of the United States Air Force

Specifications

Data from Jane's All The World's Aircraft 2003–2004^[5]

General characteristics

- **Crew:** 1
- **Length:** 46 ft 8 in (14.22 m)
- **Wingspan:** 22 ft 8 in (6.91 m)
- **Height:** 9 ft 3 in (2.82 m)
- **Wing area:** 220 sq ft (20.4 m²) (estimated)
- **Max takeoff weight:** 7,400 lb (3,356 kg)
- **Powerplant:** 1 × Pratt & Whitney Canada JT15D-5C, 3,190 lbf (14.2 kN) thrust

Performance

- **Maximum speed:** 260 kn (299 mph, 482 km/h)
- **Service ceiling:** 20,000 ft (6,100 m)

See also

Related development

- Boeing X-45

Aircraft of comparable role, configuration, and era

- Lockheed Have Blue
- Northrop Tacit Blue
- McDonnell Douglas X-36
- Boeing Model 853-21 Quiet Bird



Bird of Prey exhibit at the National Museum of the United States Air Force

References

1. "Boeing Unveils Bird of Prey Stealth Technology Demonstrator" (http://www.boeing.com/news/releases/2002/q4/nr_021018m.html) Archived (https://web.archive.org/web/20041010171318/http://www.boeing.com/news/releases/2002/q4/nr_021018m.html) October 10, 2004, at the Wayback Machine. Boeing, October 18, 2002.
2. USAF Museum literature
3. "Now you see it, now you won't: Boeing lifts the veil on stealthy Bird of Prey". *Jane's International Defence Review*, December 2, 2002.
4. Suci, Peter (May 1, 2020). "Boeing's Stealth YF-118G 'Bird of Prey' Was Straight Out of Science Fiction" (<https://nationalinterest.org/blog/buzz/boeings-stealth-yf-118g-bird-prey-was-straight-out-science-fiction-149656>). *The National Interest*. Archived (<https://web.archive.org/web/20210102041836/https://nationalinterest.org/blog/buzz/boeings-stealth-yf-118g-bird-prey-was-straight-out-science-fiction-149656>) from the original on January 2, 2021. Retrieved January 29, 2021.
5. Jackson 2003, p. 580.
 - Jackson, Paul. *Jane's All The World's Aircraft 2003–2004*. Coulsdon, UK: Jane's Information Group, 2003. ISBN 0-7106-2537-5.

External links

- Boeing Bird of Prey (<https://www.nationalmuseum.af.mil/Visit/Museum-Exhibits/Fact-Sheets/Display/Article/196041/boeing-bird-of-prey/>) at the National Museum of the United States Air Force.
- YF-118G Bird Of Prey: The 'Stealth Fighter' Lost To History (<https://www.19fortyfive.com/2022/07/yf-118g-bird-of-prey-the-stealth-fighter-lost-to-history/>) Archived (<https://web.archive.org/web/20220705191716/https://www.19fortyfive.com/2022/07/yf-118g-bird-of-prey-the-stealth-fighter-lost-to-history/>) 2022-07-05 at the Wayback Machine, 1945.

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